

A Machine Reads the Hopper Ledgers: Constrained Transcription, Mechanical TEI Export, and Checkability as the Editorial Condition

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Abstract

This article reports an experiment in the transcription and encoding of an artist's record books by a large language model under editorial constraints, using the six ledgers of Edward and Josephine Hopper at the Whitney Museum of American Art (504 digitised sheets, 1913–1967) and four of Josephine Hopper's notebooks from the Sanborn Hopper Archive as its corpus. The model (Claude Opus 5 and Claude Fable 5.1) transcribed the sheets from the museum's published photographs into a deliberately restricted LaTeX whose macros correspond one to one with elements of a TEI P5 customisation; a mechanical conversion produces TEI files that are validated against the customisation in continuous integration; scripts derive from the transcriptions, under published rules, an account of prices, materials, formats and dated days; and an interface built for the purpose places the reading beside the photograph so that any sentence can be checked against the hand it was read from. The article describes the corpus, the transcription rules, the encoding and its schema, the derived data and the interface, and then discusses the limits of the method: the reading remains a first pass unchecked by a person, an image-alignment pass on one volume reports seventy-four leaves with at least one line the transcription lacks, and the model cannot see the colour of ink that carries the cash book's meaning. The contribution claimed is not a finding about Hopper but an editorial condition: a machine reading that is fully cited, fully derived and fully exposed to correction. The use of generative AI in the transcription and in the drafting of this article is declared.

Keywords. TEI; artist's ledgers; large language models; transcription; critical apparatus; Edward Hopper; reproducibility.

1 Introduction

Edward Hopper (1882–1967) recorded his finished work in a set of ledgers that his wife, the painter Josephine Nivison Hopper (1883–1968), kept from their marriage in 1924 until his death. Six volumes survive at the Whitney Museum of American Art, accessioned as 96.208 to 96.213, and the museum has digitised and published every sheet with a catalogue descriptor. The volumes are heterogeneous. Three are work books in which Edward Hopper drew a small ink record of each finished picture with its title and size, and Josephine Hopper wrote everything else: the paint used, a description, the buyer, the price, the dealer's commission, the exhibitions and their verdicts, and the date the cheque cleared. One is a pocket cash book kept without a gap from 15 November 1913 to 23 March 1967, in a hand that predates the

marriage by eleven years and that the present edition does not attribute. One is a book of drawings and etching stock-takes, and one is a cash book indexed by dealer. Beside them, the Whitney's Sanborn Hopper Archive holds some ninety notebooks of Josephine Hopper's, four of which have been digitised: a diary of the marriage as it bore on her own painting, two working files from the 1947 fight against New York University's attempt to evict the tenants of 3 Washington Square North, and a two-ring memorandum book of lists.

The ledgers have been used before. Gail Levin's catalogue raisonné was written from the record books and her biography from the diaries (Levin 1995a, 1995b); Deborah Lyons published the record books in 1997 (Lyons 1997); Madeleine Larson's typescript of the Provincetown diaries exists in five parts. What has not existed is a transcription of the digitised sheets that can be placed beside the photograph of each sheet and checked against it, with an apparatus that states what was read, what was doubted, what could not be read and whose hand wrote it. This article reports an experiment in producing such a transcription with a large language model as the first transcriber, under constraints designed to make its output checkable rather than trustworthy.

The objective is methodological. The Hopper corpus was chosen because it is the right size for a test, because the museum serves its photographs openly at a resolution at which Josephine Hopper's figures are legible, and because a ledger is a document in which errors are measurable: a price either adds up or it does not. The article therefore presents the corpus (Section 3), the transcription method and its rules (Section 4), the TEI encoding and the customisation that validates it (Section 5), the interface and the derived data that make the reading reproducible (Section 6), and a discussion of what the method permits and what it does not (Section 7). A longer critical note on the content of the ledgers, drafted by the same model from the same transcriptions, accompanies the edition and is cited here where a finding illustrates a point of method (Hua 2026).

2 Related work

Three bodies of work bear on this experiment: the encoding of financial and account records in TEI, the transcription of handwritten historical documents by large language models, and the evaluation of such models as producers of TEI markup.

The TEI encoding of account books has a small but explicit literature. Tomasek and Bauman (2013), writing in this journal, proposed a stand-off model of financial records, the «transactionography», in which transfers between parties are represented as data linked by pointers to the transcribed document rather than embedded in its markup, so that a transaction can be encoded once even where the document records it in several places. That distinction between the document and the transaction is the one the present edition adopts, with a different division of labour: the transcription encodes the document and nothing but the document, and every transaction is recovered from it by a script whose rules are published and whose output is regenerated on each build (Section 6). The present corpus is also unlike a double-entry book in that its writer kept no consistent form; the reader that recovers transactions from Book IV has to change its rule four times across the volume's fifty-four years, and Section 4 describes why.

The transcription of handwriting by large language models has moved quickly. Humphries and colleagues (2025) report that, on a corpus of eighteenth- and nineteenth-century English manuscripts, general-purpose language models achieved character error rates of about six to

seven per cent, an improvement on specialised handwritten text recognition software; Crosilla, Klic and Colavizza (2025) benchmark several models on the same task and discuss the conditions under which the models’ linguistic priors help and hinder. The present experiment differs from that work in its object and in its measure. Its object is not running prose but ruled tables of figures, names and abbreviations, where a language model’s prior is of little use and may be a liability, since a plausible name is not a correct one and a rounded figure is a false transaction; and its measure is not an error rate against a ground truth, which does not exist for these sheets, but the reconciliation of the transcribed figures with the sums the writer herself pencilled at the foot of the leaves.

The evaluation of language models as producers of TEI is more recent. Strutz (2026) proposes a multi-dimensional framework for assessing model-generated TEI, from syntactic validity to semantic accuracy, on historical correspondence. The present edition takes a different position on the question that framework addresses: it does not ask the model to produce TEI at all. The model writes a restricted transcription language of fourteen macros, and the TEI is produced from it mechanically, so that the validity of the encoding is a property of the converter and not of the model, and a difference between the two files is a bug rather than a judgement. Section 5 sets out the reasoning and Section 7 its cost.

3 The corpus

3.1 The six ledgers

The Whitney’s six volumes are listed in Table 1. Three characteristics of the corpus shaped the method.

Table 1: The six ledgers as the Whitney Museum identifies them, with the number of digitised sheets and the state of the transcription on 13 September 2026.

Volume	Accession	Dates	Contents	Sheets
Book I	96.208	1913–1963	Etchings, one plate to a leaf; running lists from leaf 45	117
Book II	96.209	1907–1962	Paintings, one to an opening, with record sketches	72
Book III	96.210	1924–1967	Late work; lists at the front; memoranda	90
Book IV	96.211	1913–1967	Pocket cash book, continuous	161
Book V	96.212	1953–1963	Drawings; etching stock-takes	16
Dealers/Etchings	96.213	1921–1951	Prints by dealer; leaves numbered from 51	48

First, three numbering systems overlap on every sheet and only one of them is an address. The Hoppers wrote leaf numbers on the paper; the museum’s digitisation counts photographs, covers and versos included, so that it agrees with the leaf numbers nowhere; and the museum’s ResourceSpace holds a resource reference that is in no order but that names exactly one image. «Leaf 58» names six different things across the six volumes and three photographs within Book I alone. A citation from the edition therefore prints both the leaf, which is what a reader of the photograph can see, and the reference, which is the only thing the museum can be asked about.

Second, the volumes were written by two people and annotated by more. The inside front cover of Book I states the division: «Recorded by Jo N Hopper (Mrs. Edward Hopper) at time each work is finished, or before it leaves studio. Drawings in the 3 books done by Edward Hopper.» The size in the title line of each work is the one figure Edward Hopper wrote; everything else in the work books is hers. Book IV is in a hand the edition cannot

place. A transcription that flattened the hands would destroy the archive’s chief documentary value, and the encoding of hands is accordingly the one element of the apparatus that has no counterpart in ordinary manuscript transcription and that the method treats as mandatory.

Third, the colour of ink carries information. In Book IV, from leaf 7 onward, charges are written in black, receipts in red, and the sums the writer made herself in pencil; on the etching leaves of Book I, the writer states on the index leaf that ink marks the date a cheque was received and pencil the date a picture was sold. A transcription in one typeface loses this unless it records the colour, and a language model reading a photograph does not reliably distinguish it (Section 7).

3.2 The four notebooks

The four notebooks digitised by the Whitney are listed in Table 2. They are not of one kind. *Garrulities + Grouch* is a diary of twenty-two dated entries between 1927 and 1959, with the dates in the margin and the entries not in chronological order. *Re: 3 Wash Sq* is a working file of drafts and a chronicle, in which the drafts were written from the front of the pad and the chronicle from the back, so that the two meet in the middle and the chronicle runs backwards through the photographs. *Battle of Wash Sq.* is a memorandum book of eight sheets. The black two-ring notebook is a book of lists, whose dated content runs from 1937 to 1964 and whose subject changes every page or two. Their encoding required one element the ledgers do not: a dated entry, with the writer’s date and the transcriber’s assigned date kept apart.

Table 2: The four notebooks, Sanborn Hopper Archive, Series IV, Subseries A, with the Whitney’s dates and the dates of their contents as read.

Notebook	Whitney’s date	Contents as read	Sheets
Garrulities + Grouch	1924–1950	1927–1959	42
Re: 3 Wash Sq	1947	January–June 1947	51
Battle of Wash Sq. (EJHA.1597)	1947	1947–1955	8
Black two-ring notebook	c. 1952–1961	1937–1964	68

3.3 Provenance and access

The three work books were the gift of Lloyd Goodrich, who organised the Whitney’s Hopper retrospectives of 1950 and 1964; Book IV and the dealers’ book were purchases. The notebooks came with the Sanborn Hopper Archive in 2017. Twenty-two further diaries of 1933 to 1956 are at the Provincetown Art Association and Museum, which publishes Madeleine Larson’s typescript of them; those are not transcribed here and are used only through the typescript, cited by part and page.

The edition stores no photograph. The Whitney’s download endpoint answers cross-origin requests and honours byte ranges; the interface points an image element at the museum’s file, fetched as it is looked at, and the repository refuses image files. The inventory of sheets was harvested by a person in a browser, because the museum’s listing pages sit behind a bot-check that the edition does not circumvent, and the harvest is committed as text so that it can be reviewed in a diff.

4 The transcription method

4.1 The unit of work and the model

A batch is twelve sheets, read in one fresh model context, and written as a single LaTeX file whose header records the model, the date, and what the batch found difficult. The skill that performs the reading accepts three models, Claude Fable 5.1, Claude Fable 5 and Claude Opus 5, and refuses any other, on the ground that a pass made on a model nobody chose produces a file whose provenance is an accident. Every file records the model that read its sheets, so that provenance is a fact about the file and not about the session. Between 5 and 13 September 2026, Opus 5 read most of the ledgers and the first sittings of the notebooks; Fable 5.1 read the last sittings of the black notebook and the Battle of Wash Sq. Where the rule of one context per batch was not kept, the header says so: batches 3 and 4 of Book I shared a context.

4.2 The rules

The governing rule is stated in the edition's method and is specific to ledgers: a guessed figure is a sale that did not happen. An invented word in a letter is a bad reading; an invented price in a ledger is a transaction at a price nobody paid, to a buyer who never bought, and it will be cited, because a table looks like data in a way that prose does not. From that rule follow the others.

Nothing illegible is guessed. A passage that cannot be read is marked illegible, and the mark is never replaced by a plausible reading. A reading offered with doubt is marked doubtful. Nothing is expanded or normalised: «Inv.» stays «Inv.», «30 - 1/3» stays as written, and a name spelt four ways in the archive stays spelt four ways, because a corrected name is a claim about who bought a picture that the leaf did not make. A key to the abbreviations and notations is published separately, and each of its entries must match a transcribed leaf or the build fails. What the writer struck out is kept and marked struck, since a crossing-out is often the more informative half of an entry. Marginal additions are transcribed as marginal, with their position. The transcriber's own observations are confined to a note element that is typographically and structurally distinct from the writer's text, so that a count over the corpus can exclude it; the materials census, for example, reads only the writer's words, because the transcriber's notes discuss materials constantly and a count that read them would report a rare colourman as common because somebody remarked on its rarity.

Hands are attributed to Josephine Hopper, to Edward Hopper, to a later writer, or to nobody. *Unidentified* is a permitted answer and is, the method notes, the right one far more often than confidence suggests. The drawn lettering of the covers is unattributed; the whole of Book IV is unattributed; a pencil annotation on Book II leaf 46 in a heavier hand is attributed to nobody. Where a model attributes a hand, it does so on the letterforms and says so.

There is one edition and no second register. The project this method derives from carries a modernised text beside the diplomatic one; a ledger needs none, since its English is plain and its columns are already a table, and a second text would be a second artefact to keep in step.

4.3 What the model wrote

The model writes a restricted LaTeX defined in a preamble of fourteen macros (Table 3). The restriction is deliberate. A converter that accepted arbitrary LaTeX would silently mangle what it did not understand, and a silently mangled ledger is a table of numbers that look right; the

renderer accepts only the subset and fails loudly outside it, naming the file, the line and the construct. It also refuses a resource reference that is not in the harvested inventory and a file that omits the rights notice.

Table 3: The transcription macros and the TEI elements the converter maps them to.

Macro	Meaning	TEI
\sheet{ref}{leaf}	A sheet begins: the museum's reference, the leaf written on the paper	pb with @n, @facs, @xml:id
\hand{who}{...}	Whose hand: jo, edward, later, unidentified	seg with @hand, and handNote
\ill{}	Illegible; never guessed	gap reason="illegible"
\uncertain{...}	A reading offered and flagged	unclear
\struck{...}	Struck out in the book	del
\add{...}	An editorial addition	supplied
\note{...}	The transcriber's note	note type="editorial"
\marginal{...}	The writer's marginal note	note place="margin"
\work{...}	A work heading, title and size	div type="work" with head
\sketch{...}	Edward Hopper's ink record drawing stands here	figure type="record-sketch"
\clipping{...}	Something printed and pasted to the leaf	figure type="clipping"
\ink{colour}{...}	The colour a cell is written in, where it means something	hi rend="ink-red" etc.
\entry{as written}{ISO}	A diary entry begins; the writer's date and the assigned date kept apart	div type="entry" with head/date@when
ledgertable	The ruled columns	table, row, cell

The choice of LaTeX rather than XML as the model's output language was practical and is discussed in Section 7. Its consequence for encoding is that the model never writes an element name, an attribute or a namespace, and cannot therefore produce ill-formed or invalid markup; what it can do is misapply a macro, and the converter's checks and the schema's closed value lists are designed to catch the misapplications that have a structural signature.

4.4 A transcribed leaf

The following is the writer's description of Nighthawks, from Book II leaf 95, as the model read it, given here in the TEI that the converter produces from the transcription (the sheet is ref 17982). The struck word, the marginal additions with their positions, the hand and the transcriber's note are all present, and the museum's reference is carried in the page break.

```
<pb n="95" facs="https://resourcespace.whitney.org/pages/view.php?ref=17982" xml:id="sheet-17982"/>
<p><seg hand="#jo">1942</seg></p>
<figure type="record-sketch"><head>Ink record sketch by Edward Hopper</head></figure>
<div type="work">
<head type="work"><« Night Hawks » 33 x 60</head>
<p><seg hand="#jo">Finished Jan 21, 1942</seg></p>
<p><seg hand="#jo">Blockx and Winsor & Newton colors, W & N. Zinc white, poppy oil. English linen, domestic priming.</seg></p>
<p><seg hand="#jo">Night + brilliant interior of cheap restaurant. Bright items; cheery wood counter + tops of surrounding stools; [...] Sign across top of <del>shop</del> dark - Phillies 5¢ Cigar, picture of cigar.</seg></p>
<note place="margin">curving - above « window », with a caret beneath the line</note>
<note place="margin">restaurant - above the struck « shop »</note>
```

```

<note type="editorial" resp="#editor">this is the other side of the transaction batch 2
transcribed at leaf 25, where the Art Institute buys Compartment C in 1938 and
returns it in 1942 in part payment for Night Hawks. [...]</note>
</div>

```

5 The TEI encoding

5.1 Why the TEI is derived and not written

The TEI files are produced by a script from the transcriptions alone. The script re-reads no sheet, and the method requires that it must not: the macros map one to one onto TEI elements, the conversion is mechanical, and a difference between a transcription and its TEI is by construction a defect in the converter and not an editorial decision. This is the reverse of the usual division, in which the TEI is the source of record and a display is derived from it, and it was chosen for three reasons.

The first is the model. A language model can be asked to write TEI and will do so fluently, and its fluency is the problem: it will produce well-formed markup whose structure is plausible and whose attribute values are invented, and the failure has no structural signature. A fourteen-macro language has a small surface, each macro answers one editorial question, and a macro used wrongly is visible as a wrong answer to that question rather than as a subtly wrong element.

The second is the checker. The transcription is read by people beside the photograph, in a rendering the converter also produces from the same file, and the rendering has to be legible to a reader who knows nothing of XML. A LaTeX source with fourteen macros is also readable in itself, in the way a diplomatic transcription in a printed edition is, and can be corrected by a person with a text editor.

The third is validation. Because the TEI is derived, it can be validated on every build against a customisation that declares exactly the elements the converter emits, and a validation failure identifies a converter bug or a macro misuse without any ambiguity about whose judgement is at fault.

5.2 The customisation

The customisation is an ODD, `hopper.odd`, served at the URL every exported file names, from which both the schema and its documentation derive. It declares fifty-five elements, and the fifty-five are not a sample of TEI's six hundred but the exact reach of the conversion: every element exists because a macro maps onto it. The modules and elements included are shown in Table 4. Five attribute value lists are declared closed, so that a `@type` on a division, a figure, a note, a term or a rendition value on a highlighted cell is drawn from a list and not invented per file, and the validator refuses any other value. The value lists are: division types (`leaf`, `work`, `entry`, `part`, `editorial`); figure types (`record-sketch`, `clipping`); note types (`editorial`); the keyword facets (`person`, `place`, `collection`, `dealer`, `society`, `publication`, `prize`, `medium`, `client`, `subject`, `feature`, `hopper`); and renditions (`bold`, `italic`, `mono`, `ink-red`, `ink-pencil`, `ink-blue`, `rule-above`).

The header of each file carries the provenance that the method requires. The two authors are named in the order the inside cover gives them; the responsibility statement names the model that read the sheets and the date, and states that the pass is a first machine pass

Table 4: The modules and elements the ODD includes; nothing else validates.

Module	Elements included
textstructure	TEI, body, div, text
core	author, bibl, date, del, gap, head, hi, item, lb, list, name, note, p, pb, publisher, resp, respStmt, term, title, unclear
header	availability, catDesc, category, classDecl, edition, editionStmt, editorialDecl, encodingDesc, fileDesc, handNote, idno, keywords, profileDesc, publicationStmt, sourceDesc, taxonomy, teiHeader, textClass, titleStmt
msdescription	history, institution, msDesc, msIdentifier, origDate, origin
figures	cell, figure, row, table
transcr	handNotes, supplied
linking	seg

unchecked by a person; the availability statement records that the transcribed text is in copyright and that the file is an unauthorised working document; the manuscript description carries the museum's accession number and the volume's dates; the hand notes declare the four hands; and the text classification carries the keywords, typed by facet.

```
<titleStmt>
  <title>Artist's ledger – Book II, batch 6</title>
  <author>Josephine Nivison Hopper</author>
  <author>Edward Hopper</author>
  <respStmt>
    <resp>Transcription (first machine pass, unchecked against the sheets by a person)</resp>
    <name>Opus 5 (claude-opus-5), 2026-09-06</name>
  </respStmt>
</titleStmt>
[...]
<handNotes>
  <handNote xml:id="edward"><p>Edward Hopper</p></handNote>
  <handNote xml:id="jo"><p>Josephine Nivison Hopper</p></handNote>
  <handNote xml:id="later"><p>A later hand, after 1967</p></handNote>
  <handNote xml:id="unidentified"><p>Unidentified</p></handNote>
</handNotes>
```

5.3 Tables, ink, and the cash book

The ruled tables are encoded as tables, one row per ruled line and one cell per column, with an empty label row where the leaf heads no column. The colour of ink, where the pass that reads it has run, is encoded as a rendition on the highlighted cell; a rule the writer drew across the money column above a row is a rendition on the row. The following rows from Book IV leaf 121 show a charge in black, a receipt in red, a sum in pencil and a ruled subtotal, and one receipt whose day and word could not be read.

```
<row><cell>June 12</cell><cell>Frank K. M. Rehn</cell><cell></cell><cell></cell></row>
<row><cell></cell><cell>Sattillo Roof Tops w.c</cell><cell>750</cell><cell>00</cell></row>
<row><cell></cell><cell>less commission</cell><cell>250</cell><cell>00</cell></row>
<row rend="rule-above"><cell><hi rend="ink-red">" <unclear>12</unclear></hi></cell>
  <cell><hi rend="ink-red"><unclear>rec'd</unclear> <gap reason="illegible"/></hi></cell>
  <cell>500</cell><cell>00</cell></row>
<row><cell><hi rend="ink-red">" 12</hi></cell><cell><hi rend="ink-red">rec'd by check</hi></cell>
  <cell><hi rend="ink-red">500</hi></cell><cell><hi rend="ink-red">00</hi></cell></row>
```

```
[...]
<row><cell></cell><cell><hi rend="ink-pencil">1800 + 560</hi></cell>
  <cell><hi rend="ink-pencil">53</hi></cell><cell></cell></row>
<row rend="rule-above"><cell></cell><cell></cell>
  <cell><hi rend="ink-pencil">2360</hi></cell><cell><hi rend="ink-pencil">53</hi></cell></row>
```

The two red rows are one receipt: the writer wrote it across the row that carried the subtotal, rubbed it out, and wrote it again below, and the transcriber's note on the leaf says so. The accounts reader counts it once. This case, which recurs on nine leaves of the volume, is the clearest instance of why the transaction is recovered from the document by a rule rather than encoded in it: the document records the receipt twice, and the fact that it is one receipt is an inference.

5.4 The notebooks

A diary entry is a division of type entry whose head carries the writer's date as written and the assigned date in the @when attribute. The two are kept apart because the assignment is the one editorial claim an entry carries: *Garrulities* has entries whose day is a blot and whose year is inferred from the run of dates around them, and the encoding must not present an inference as a reading. The following is the black notebook's page 64, the one sentence of Edward Hopper's stated purpose that the ten books preserve, with the writer's own recasting of it below and the transcriber's note distinguishing the two.

```
<pb n="64" facs="https://resourcespace.whitney.org/pages/view.php?ref=98119" xml:id="sheet
-98119"/>
<div type="entry">
<head><date when="1946-01">Jan 1946</date></head>
<p><seg hand="#jo">Jan 1946. <hi rend="italic">Quoting E. H.</hi> <lb/>
Cruelly pinned down to definition <lb/>
of his purpose generally, I extract: <lb/>
" the arrestation of a moment <lb/>
in time - with the most acute <lb/>
realization.</seg></p>
<p><seg hand="#jo"> to arrest <lb/>
a moment in time <lb/>
<del>with</del> that moment <lb/>
most acutely realized. "</seg></p>
<note type="editorial" resp="#editor">both blocks are enclosed in red crayon - the first
from « I extract » down, the second entire - and the second is her own recasting of the
first, not a second saying of his.</note>
```

5.5 Validation

The converter's output is validated in continuous integration against the schema derived from the ODD, and the workflow fails on any invalid file. The validator exists because of a defect it found: for as long as the export was checked only for well-formedness, every file in the corpus was well-formed XML and invalid TEI, since the converter had placed a head at each leaf inside a single division per batch, and TEI places a head only at the start of the division it heads. Nothing had said so because nothing had asked. The lesson is general and is recorded in the validator's own header: a deposit of unvalidated TEI is a deposit of XML.

6 Interface and reproducibility

6.1 The reading beside the photograph

The interface at <https://hopper.commutator.io> was built for this project and does one thing that no general tool did: it renders the transcription in a left pane and the Whitney's photograph of the sheet in a right pane, and scrolling the transcription past a page-break element turns the photograph to that sheet. Every doubtful figure can be checked against the hand it was read from without a second window. A Cite control on each sheet builds three citations from what the page already knows, a sheet, a leaf and a batch, each naming the museum's object clause and the edition's reading clause separately, and offers only those that exist: a sheet the Hoppers did not number gets no leaf citation, because inventing a number for the sake of a tidy citation would invent a fact about the object. The citation names the model and the date of the reading, because a citation of this edition cites a reading and not a fact, and a reading that could not be dated could not be superseded.

The reading views for the notebooks work the same way and are addressed by the museum's reference rather than by a page number, since an opening photographed whole carries two page numbers and a cover none.

6.2 Derived data, and the rule that a build must fail

Everything the interface shows beyond the transcription is derived from the transcription by a script whose header states what it counts and what it therefore is not, and the continuous integration workflow regenerates each derived file and fails if the committed copy differs. The manifest of what is transcribed is observed from the files, never declared, so that a progress figure cannot claim work that was not done; the only declared states, *checked* and *skipped*, are held in a separate file that is empty. The citation metadata is generated from a census of the files, on the ground that a hand-written sheet count is wrong from the next batch onward. The glossary of abbreviations and notations is hand-declared, but each entry carries a pattern that must match a transcribed leaf, and the search runs over the writer's words and never over a transcriber's note, so that an entry cannot cite as evidence the sentence written because of it.

The accounts reader recovers transactions from the ruled rows. It counts a row only where the leaf puts a price in one cell and a date in another, because the column a date stands in is what says whether it dates the sale or the cheque; on the etching leaves of Book I the first column dates the exhibition, so those rows are set aside rather than dated wrongly. Where a row states a price, a rate and a receipt, the receipt is checked against the price less the commission; of ninety-five such rows, ninety agree to the cent, and the five that do not are reported and not corrected. For Book IV the reader has a second set of rules, set out period by period in a method note, because the writer's form changes four times across the volume: items are the charge; a bill counts only where nothing was itemised since the last settlement; a bare figure is a subtotal where the arithmetic of the lines above it says so, since the leaf rules a line above a subtotal and the transcription does not record the rule; a «less» line is a deduction whichever thing it names; a title charged more than once with a payment on account is one picture at the largest figure. The check on the reader is the book's own pencil totals, and the reader's receipts reconcile to them to the cent wherever the leaf's arithmetic closes.

Two of the reader's rules were found by bugs, and the method note records them: an unanchored match on the word *bill* took a story title, «Wild Bill in Deadwood Gulch», for an invoice and lost a drawing of 1915; a header rule read «May 25th Initial Letters» as a new

client and counted a bill of 225 twice. Both were caught by comparing per-year totals with leaves read by hand. A faithful transcription can surface a reader bug without containing one.

6.3 The rules, stated as proofs

A reader whose rules are written in prose and applied by a script can drift from them unnoticed, and a total can reconcile by accident. The derived accounts are therefore also restated in the Lean 4 theorem prover, in a directory of the repository that depends on nothing but Lean, so that the continuous integration build fails if the script and its stated rules disagree. The formalisation was written by Claude Opus 5 under the author's direction, on 14 September 2026. Every figure in the ledgers is dollars and cents and every commission a third, a quarter, a tenth or fifteen per cent, so money is represented as integers in sixtieths of a cent, where all of them are exact. General lemmas are proved for every input by decision procedures for linear arithmetic; the data is checked by evaluation in the kernel (`decide +kernel`) and never by compiled code; and warnings are errors, so an unfinished proof fails the build.

Two certificates are proved. The first concerns the accounts as published: each activity's years are recomputed from its rows, and the identity that what accrued equals what was collected plus what is outstanding plus what was not recorded is derived from the rows rather than read off the totals; the reader's two-cent tolerance is proved to accept every figure written to the cent after a truncated third and to reject a slip of a third of a dollar; the ninety-five checkable rows are proved to yield ninety agreements and exactly the five disagreements the reader reports; and each hand-declared pencil total is proved equal to the reader's receipts for that year plus the declared omissions. The second concerns the Book IV reader. Its four passes are stated as Lean functions, and the properties the method note asserts in prose are proved of them for every input: a bill is charged only where nothing was itemised since the last settlement, so no commission is counted twice; a «less» line is never a charge; a bare figure is a subtotal exactly when the answering line or the arithmetic says so; a picture paid in instalments is charged once; the year does not carry across a gap between batches. The kernel then evaluates the functions over all 3,281 transcribed rows of the volume and proves that each row is decided exactly once, that the Lean reader and the JavaScript reader agree on every row's kind, year and disposition, and that the published counters are counts of those decisions. The JavaScript exports, for each row, the lexical features it recognised and the decisions it took; Lean recomputes the decisions from the features and proves equality row by row. Altering a single disposition in the exported data causes the build to fail.

The exercise refined one rule, since the proof required the exception that a «less» line which is also a ditto immediately under a receipt reads as that receipt, and it turned a known gap into a theorem: the absence of 1921 from the cash book follows from a year line on leaf 53 that reads 1920 over entries of 1921, and the corresponding theorem is written to fail once that line is corrected at source. The limits are those of the method as a whole. The certificates say nothing about a reading, and a figure misread consistently in the rows and in the sums would pass; the regular expressions that recognise a row's words are inputs, not proved; and what is proved is that the script agrees with its stated rules, not that the rules are the right interpretation of the writer's practice. The certificates are checked on every push by the workflow that builds the site, and an equivalent GitLab CI configuration is provided.

6.4 What the data shows, as an illustration of the condition

Figures 1 to 4 reproduce three of the interface's drawings from the derived files, and are given here not for their findings but to show what a fully derived figure looks like: each names its source file and that file's generation date, and each is regenerable by anyone with a clone of the repository.

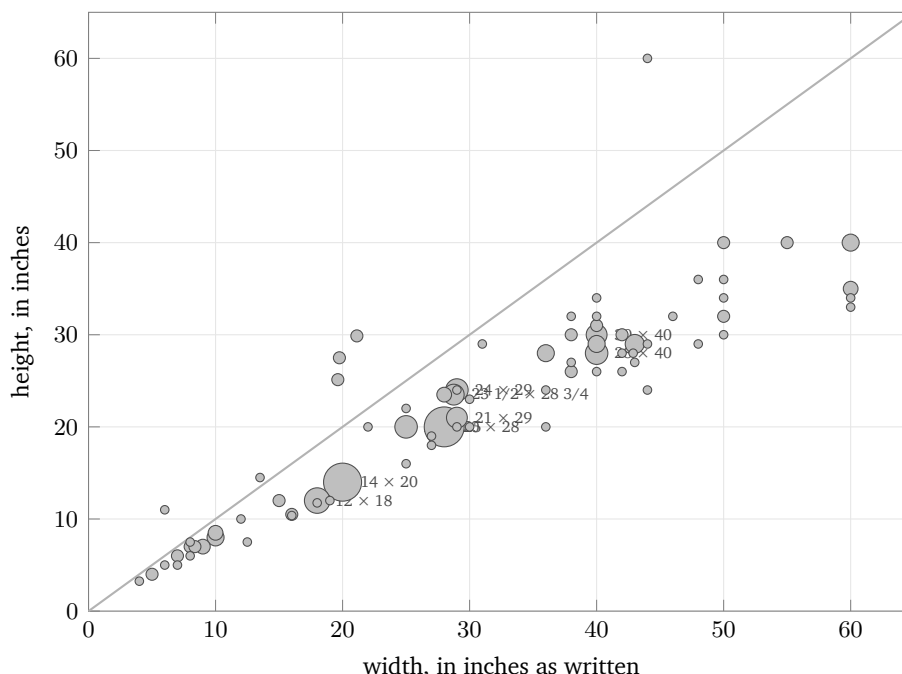


Figure 1: Every format, plotted: width across, height up, one dot to a format and its area standing for how many works were made at it, 78 formats from 199 work headings that state a size. The diagonal is where a picture would be square; 190 of the 199 works lie below it, 9 above, none on it. The etching plates are in the corner, the watercolour sheets through the middle, the late oils at the top right. After the Technique tab of hopper.commutator.io. Drawn by Claude Fable 5.1, 13 September 2026, from `formats.json` (undated).

The format census reads only the size in the title line, in Edward Hopper's hand: 199 of 270 work headings state one, they make 78 formats, 72 of them wider than tall and none square. The materials census reads only the paint formula in Josephine Hopper's hand, counted once per work and never from a transcriber's note: 77 headings carry one, and the succession they state, Rembrandt colours and zinc white in poppy oil in 1932, Winsor & Newton and flake white in linseed from the middle 1940s, the canvas turning from National to Herga and the priming from double to single across the 1950s, is dated by the leaves themselves for 43 works and by nothing else. The money is given three ways, from the work books and from the cash book, because the two are different records and should not be added. The days on which the books record a transaction number 1,204 over fifty-four years, with a sixth of a weekday's days on Sundays and a hollow every summer, which is the Cape; the drawing is captioned to say that a dense month is a month of transactions and not of painting, because a grid of squares reads as productivity and here that is backwards.

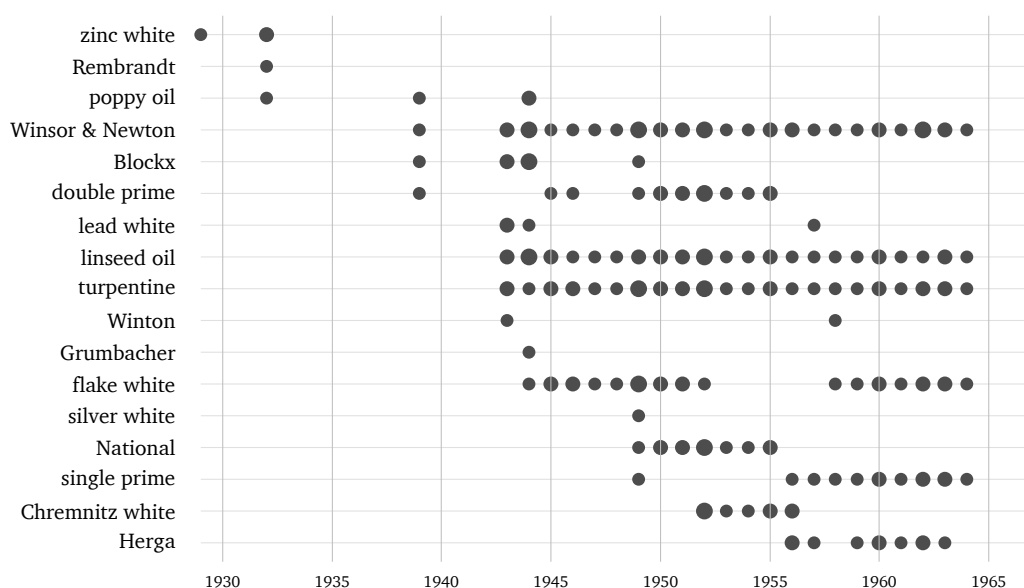


Figure 2: A succession, not a preference: each material against the years the leaves date it to, earliest first, one dot to a year and its size standing for how many works that year names it. Rembrandt colours and zinc white in poppy oil at the start; Winsor & Newton and flake white in linseed oil and turpentine from the middle 1940s; the canvas turning from National to Herga and the priming from double to single across the 1950s. Only the 43 works dated by the leaf itself are plotted. After the Technique tab. Drawn by Claude Fable 5.1, 13 September 2026, from `materials.json` (undated).

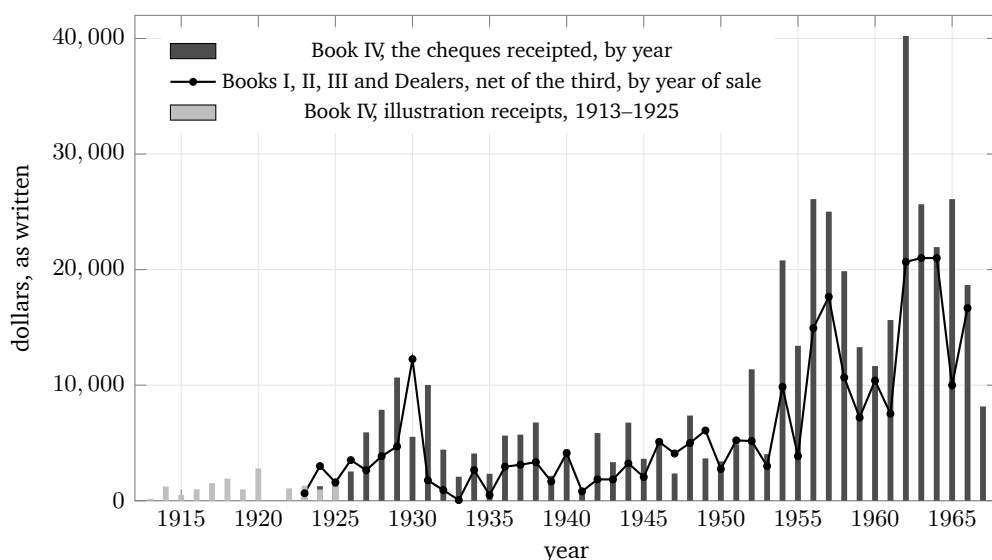


Figure 3: The money, three ways: what the pocket book receipts each year from 1920 (dark bars), what the work books state as the net of sales dated in that year (line), and the illustration receipts of the first decade (pale bars). The two art series are not the same number and should not be: one is the dealer's cheques as they arrived, the other the studio's record of what left. Every figure is a floor, read from the transcribed rows under the rules of the accounts chapter. After the Accounts tab of `hopper.commutator.io`. Drawn by Claude Fable 5.1, 13 September 2026, from `accounts.json` (2026-09-11).



Figure 4: The days the six books record, one row to a year from 1913 to 1967 and one cell to a month, the darker the more days that month carry an entry: 1204 days in all, from 2146 dated rows. A dense cell is a month of transactions, not of painting; the hollow every summer is the Cape, with the dealers behind them in New York. By day of the week: Sun 34, Mon 197, Tue 197, Wed 179, Thu 215, Fri 207, Sat 175, Sunday a sixth of a weekday. After the calendar of the Timeline tab, which draws every day; this draws every month. Drawn by Claude Fable 5.1, 13 September 2026, from `streak.json` (2026-09-13).

7 Discussion

7.1 The reading against the image

The transcription is a first pass and has not been checked by a person. For this article, the two quotations on which its accompanying critical note leans hardest were compared with the museum's photographs by the model itself on 13 September 2026, at the published resolution of 1292 by 2000 pixels; Table 5 gives the result cell by cell. Both readings hold in every word, and the comparison found one omission: a small mark before the struck word on the notebook page, which the transcription does not record. The comparison is reported for what it is. It was made by the same kind of reader that made the transcription, it covers two sheets of 543, and it establishes only that a second reading agreed with the first. The edition reserves the word *checked* for a comparison made by a person, and the comparison is recorded in the transcripts as a note that says who made it and that it is not that.

Table 5: Transcription against image, for the two sheets compared on 13 September 2026. Sheet refs 98119 (black notebook, page 64) and 16838 (Book III, leaf 19).

Sheet, place	As transcribed	As seen in the photograph
98119, head	«Jan 1946. <i>Quoting E. H.</i> »	Agrees; the second phrase underlined
98119, l. 2–3	«Cruelly pinned down to definition / of his purpose generally, I extract: »	Agrees
98119, l. 4–6	«" the arrestation of a moment / in time - with the most acute / realization. »	Agrees; «acute» underlined; block enclosed in red crayon from «I extract» down
98119, block 2	«to arrest / a moment in time / with that moment / most acutely realized. " »	Agrees; «acutely» underlined; a small cross before the struck «with» is not recorded
16838, title	«October on Cape Cod. 26 x 42. Finished on Cape Cod - in studio - Oct. 1946.»	Agrees
16838, margin	«To Rehn Gallery Nov. 26, 46» at the left	Agrees
16838, paint	«Windsor + Newton linen canvas + Colors; flake white, linseed + turpentine.»	Agrees
16838, interlined	«bearberry» above «herbage»; «low» above «clumps»; «pale» above «blue at top»	Agrees, all three
16838, close	«"Peace, quiet, no birds sing." "Some day you will be quiet too" - E. H.'s favorite Goethe. (strictly off the record) »	Agrees
16838, sale	«3000 - 1/3 = 2000 June 14, 1956.»	Agrees

A more systematic comparison exists for one volume and shows the limit of the reading. A pass that aligns each transcribed row of Book IV with the written lines of its photograph, in order to read the ink colour the model cannot see, reports seventy-four leaves, about half the volume, with at least one written line the transcription does not have. Many are the alignment's own noise, and each must be looked at; at least one, an entry of 26 February 1917 on leaf 21, was a real omission. The consequence is bounded and can be stated: a missing line changes a sum, so the cash-book figures are floors in a second sense, and no such pass has run on the other volumes or the notebooks, about whose omissions nothing is therefore known.

7.2 Limits of the model as a transcriber and as an encoder

The experiment identified five limits, none of them a matter of error rate.

The model does not see colour reliably. Book IV's meaning is in its ink, and fourteen batches were transcribed without recording it; the colour was recovered afterwards by an image-processing script that classifies each money cell and writes the result into the transcription, on the ground that colour is the one thing on the leaf a machine can see better than the hand it is written in. A method that depended on the language model for the ink would have produced a cash book without a status field.

The model carries context forward as a rule, and rules fail at boundaries. The year of a Book IV entry is stated once, on a line of its own, and carried down the leaf; where leaf 53's year line reads «1920» above entries of 1921, the transcription keeps the line as written and the reader carries the year, so that 1921's income sits under 1920 in every table. This is the intended behaviour of both, and it is wrong, and it is listed for correction at source rather than repaired in the derived data.

The model reads names as language. The archive spells Crowninshield three ways, Saltillo four and Keppel two, and the model, left to itself, would settle them. The rule that nothing is normalised holds it to the leaf, and the glossary records the variants; but the discipline is the rule's and not the model's, and a reader who relaxed it would get a tidier and less truthful archive.

The model is fluent in TEI, and fluency is the hazard. Asked to produce markup it produces plausible markup, and a plausible attribute value is not a schema-valid one. The decision not to let the model write TEI at all, and to derive the TEI mechanically from a fourteen-macro language it can hardly misuse, is the central encoding decision of the edition. Its cost is that the encoding is only as expressive as the macro set, and the macro set was designed for what these ten books contain; the transactionography of Tomasek and Bauman (2013), for instance, has no place in the transcription and lives entirely in the reader.

The model is confident about hands. The attribution of a hand on the letterforms is the one judgement in the apparatus that no arithmetic can check, and the model attributes readily. The rule that *unidentified* is a permitted answer, and the decision to attribute the whole of Book IV to nobody, are the method's response; a human palaeographer with the physical volumes would do better, and the edition says so on the leaf.

7.3 Human validation

The edition distinguishes three states of a batch, and only one of them is observed from the files. *Drafted* is what a file's existence proves. *Skipped*, for a batch judged to hold nothing to transcribe, and *checked*, for a batch that a person has compared with the photographs sheet by sheet, are declared in a status file, and that file declares nothing for any batch. The reason is stated in its comment: a hand-kept status table drifts optimistic and nobody notices, so the two states no file can prove are the only two a person may write, and the site shows a transcribed batch as drafted because that is observed rather than declared.

What a human check consists of has been defined as an open question of the edition rather than assumed. The protocol under discussion is the one this article's comparison table performs in miniature: a person opens the reading beside the photograph, reads each transcribed line against the sheet, corrects the transcription at source with a note that names who checked it, and only then may the batch be declared checked; the citation of every sheet in that batch then changes to say that a person has gone through it, and only a person may declare that.

Until then, every citation of the edition names a machine pass and a date, and the address of the sheet does not move when the reading does.

7.4 What the method permits, and what it does not

The method permits a reading of a five-hundred-sheet archive to be produced in days, cited by leaf and reference, exported as valid TEI, and exposed in full to correction beside its source. It permits derived data to be regenerated from the reading under published rules and to be checked, where the writer left sums, against the writer's own arithmetic. It permits the rules by which that data is derived to be stated as functions in a theorem prover, their properties proved for every input, and the script that applies them proved to agree with them on every row. It permits the whole to be held in one repository whose every change is a dated commit, so that a citation names a version and a later reader can see what was read on the day it was cited.

It does not produce a checked transcription, and it does not claim to. It does not see ink, and it does not attribute hands with authority. It does not, in its present form, encode transactions, and it cannot produce a text more expressive than its macro language. And it does not remove the need for the person the method was designed around: the reader beside the photograph, whose correction is the only event that changes a batch's state.

8 Conclusion

This article has described an edition in which a language model is the first transcriber and the editorial apparatus is designed to make its reading checkable rather than trustworthy. The method rests on four decisions: a restricted transcription language whose macros answer one editorial question each; a mechanical derivation of TEI from that language, validated on every build against a customisation that declares exactly what the converter emits; a derivation of all counts and figures from the transcription under published rules, checked against the writer's own sums and, for the accounts, certified in Lean against the rules as stated; and an interface that places the reading beside the photograph and cites every sheet by the museum's own reference and the edition's own dated pass.

The corpus, the Hopper ledgers and notebooks, was the occasion for the experiment and is a good one, because a ledger's errors are measurable and because the museum serves the photographs openly. The critical note that accompanies the edition (Hua 2026) reads the transcription whole and is offered as a demonstration of what the method produces at first pass, its errors left in view.

What the experiment opens is a division of labour. The model reads; the scripts derive; the schema validates; and the person checks, corrects at source, and declares. The first three have been done for this corpus. The fourth has not, and the edition's own status file, which declares nothing, is the measure of how far the method has come and where it stops. Two extensions are planned: the deposit of the corpus with a persistent identifier, and the definition and execution of the human check that turns a drafted batch into a checked one. Both are tracked as issues of the repository, and the second is the one on which every claim in this article depends.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Claude Opus 5 and Claude Fable 5.1 (Anthropic), through Claude Code, in order to transcribe the digitised sheets of the corpus into the restricted transcription language described in Section 4, to compare two sheets with their photographs as reported in Section 7.1, to write the Lean formalisation of the accounts described in Section 6.3, and to draft this article and the accompanying critical note from the transcriptions and the derived data under the author's direction. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication. The transcriptions have not been checked against the photographs by a person, and the article says so wherever a claim depends on them.

Data availability

The transcriptions, the TEI export, the ODD, the scripts, the derived data, the Lean proofs, the interface and the sources of this article and of the critical note are in the public repository at <https://github.com/Commutator-I0/hopper-project>; the interface is served at <https://hopper.commutator.io>. The software and the derived data are released under CC0 1.0; this article and the critical note under CC BY 4.0. The transcribed words of Josephine and Edward Hopper are © Heirs of Josephine N. Hopper, licensed by Artists Rights Society (ARS), New York, and are quoted for criticism and study; the photographs are the Whitney Museum's and are not stored by the edition. A deposit with a persistent identifier is planned and tracked as issue 16 of the repository.

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A A complete transcribed leaf

The transcription of Book III leaf 19, October on Cape Cod (ref 16838), as the model wrote it in the restricted language, followed by the note recording the comparison with the photograph.

```
\sheet{16838}{19}

\sketch{}

\work{October on Cape Cod. \quad 26 x 42. \quad Finished on Cape Cod - in
studio - Oct. 1946.}

\marginal{To Rehn Gallery Nov. 26, 46 --- in the margin at the left of the title
and paint lines}

\hand{jo}{Windsor + Newton linen canvas + Colors; flake white, linseed +
turpentine.}

\hand{jo}{Beyond slatey grey road a strip of very pale (yellowish) sand across\
length of canvas. Low herbage digging in green. Tall grass beyond slightly\
pinkish with clumps of green. House fresh painted white, barn left dull\
Rooves of both slatey grey. Glimpse of red foundation, chimney dull red.\
Edge of locust grove L. dull green. Woods in background with green underfoot\
at front Edge. Sky wan, whitish with long strips of grey,. Some blue at top.\
"Peace, quiet, no birds sing." "Some day you will be quiet too" - E. H.'s
favorite Goethe.\
\quad (strictly off the record)}

\marginal{bearberry --- above « herbage »; low --- above « clumps »; pale ---
above « blue at top »}

\hand{jo}{Gone to Mr. + Mrs. O. W. Carpenter. Reprod. in Vogue, June 1954.
When sold?\
R. H. Bresler Co. - payment caught up in Rehn Estate -\
\quad Paid by John Clancy, new head of Rehn Gallery. 3000 - 1/3 = 2000
June 14, 1956.\
Carpenter of Milwaukee - Heavy chain, bolt Co.}

\note{the marginal at the head is at the left here where leaves 13 and 15 put
the same note at the right. « When sold? » is in a darker pencil than the line
it follows and was added to it. The two quotations closing the description are
hers to place and she places them: the first she gives no source, the second
she gives to Goethe by way of Edward, and then she puts the whole attribution
```

off the record. « favorite » is as written.}

\note{compared with the photograph at the Whitney's published size by Claude Fable 5.1 on 2026-09-13, for the critical note: the title line, the paint line, the description with its three interlineations, the two closing quotations, « E. H.'s favorite Goethe » and « (strictly off the record) », and the sale lines all read as transcribed. A machine's comparison, not the human check the edition reserves « checked » for}

B The derived tables

The tables below are regenerated from the edition's derived files on each build; those given here were generated on 13 September 2026 from 46 transcribed batches. Every figure is a floor under a stated rule.

Table 6: Book IV, the illustration decade: what was charged and what was received, by year, as the transcribed leaves state it. Accrual at the leaf's own date column; a negative balance is a cheque with no charge line behind it. There is no row for 1921 because leaf 53's year line reads 1920 above entries of 1921 and the reader keeps the year as written; that year's income is counted under 1920.

Year	Charged	Received	Balance
1913	160.00	160.00	0.00
1914	1,225.00	1,225.00	0.00
1915	500.00	500.00	0.00
1916	991.00	991.00	0.00
1917	1,509.00	1,514.00	-5.00
1918	1,556.00	1,916.00	-365.00
1919	977.00	977.00	-365.00
1920	2,950.00	2,794.00	-209.00
1922	1,075.00	1,075.00	-209.00
1923	1,279.20	1,304.20	-234.00
1924	1,015.00	980.00	-199.00
1925	1,218.67	1,218.67	-199.00
Total	14,455.87	14,654.87	-199.00

Table 7: The work books (Books I, II, III and Dealers/Etchings): sales dated in the leaf's own date column, by year. Gross is the price written; the commission is the dealer's share as the row states it; net is what was left. Only rows that put a price in one cell and a date in another are counted, so the etchings are largely absent.

Year	Sales	Gross	Commission	Net	Receipted
1923	3	970	323	647	0
1924	6	4,500	1,500	3,000	0
1925	4	2,100	510	1,590	340
1926	11	5,275	1,758	3,517	850
1927	17	3,955	1,318	2,637	1,333
1928	24	5,774	1,919	3,855	1,197
1929	52	7,040	2,342	4,698	4,197
1930	34	18,378	6,126	12,252	3,000
1931	17	2,648	883	1,765	5,600

Year	Sales	Gross	Commission	Net	Receipted
1932	3	1,380	460	920	2,400
1933	3	85	28	57	20
1934	14	3,980	1,327	2,653	800
1935	1	750	250	500	500
1936	6	4,420	1,473	2,947	267
1937	10	4,670	1,557	3,113	20
1938	11	4,990	1,648	3,342	0
1939	1	2,500	833	1,667	0
1940	14	6,182	2,061	4,121	667
1941	4	1,220	407	813	267
1942	3	2,780	927	1,853	0
1943	4	2,760	920	1,840	1,200
1944	5	4,850	1,617	3,233	0
1945	5	3,080	1,027	2,053	0
1946	6	7,650	2,550	5,100	2,267
1947	7	6,130	2,043	4,087	1,467
1948	4	7,500	2,500	5,000	1,500
1949	5	9,115	3,038	6,077	0
1950	12	4,130	1,377	2,753	1,053
1951	11	7,830	2,610	5,220	333
1952	3	7,750	2,583	5,167	2,667
1953	9	4,500	1,500	3,000	500
1954	6	14,780	4,927	9,853	0
1955	6	5,800	1,933	3,867	2,333
1956	12	22,410	7,470	14,940	5,867
1957	19	26,470	8,823	17,647	667
1958	3	16,000	5,333	10,667	1,800
1959	2	10,800	3,600	7,200	6,000
1960	4	15,600	5,200	10,400	0
1961	10	11,295	3,765	7,530	0
1962	3	31,000	10,333	20,667	3,667
1963	2	31,500	10,500	21,000	0
1964	2	31,500	10,500	21,000	0
1965	1	15,000	5,000	10,000	10,000
1966	1	25,000	8,333	16,667	0

Table 8: The thirty sales at the highest prices among the rows that name a work, as the work books write them. Gross is the price; net is after the third.

Work	Price	Net	Year	Volume
Intermission	25,000	16,667	1966	III
Sun in an Empty Room	25,000	16,667	1964	III
New York Office	16,500	11,000	1963	III
People in the Sun	15,000	10,000	1962	III
A Woman in the Sun	15,000	10,000	1963	III
East River	15,000	10,000	1965	III
Excursion into Philosophy	14,500	9,667	1962	III
« Girlie Show »	11,000	7,333	1961	II
November, Wash. Square	10,000	6,667	1960	III
Sunlight in a Cafeteria	9,000	6,000	1959	III
Hotel Window	7,000	4,667	1957	III
Western Motel	6,500	4,333	1958	III
Charleston Slum	6,500	4,333	1964	III
Morning in a City	6,000	4,000	1949	III
Four Lane Road	6,000	4,000	1957	III
Dauphinée House	5,000	3,333	1956	II
Sunlight on Brownstones	5,000	3,333	1958	III
Tables for Ladies	4,500	3,000	1930	I
Men, Women + Horses	4,500	3,000	1958	II
Sea Watchers	4,500	3,000	1956	III
South Carolina Morning	4,500	3,000	1957	III
Cape Cod Morning	4,050	2,700	1956	III
« Bootleggers »	4,000	2,667	1956	II
Rooms by the Sea	4,000	2,667	1952	III
The Martha McKeen of Wellfleet	3,500	2,333	1954	III
Portrait of Orleans	3,500	2,333	1953	III
First Row Orchestra	3,500	2,333	1954	III
Morning Sun	3,500	2,333	1954	III
Hotel by a Railroad	3,500	2,333	1954	III
City Sunlight	3,500	2,333	1955	III